Screened Poisson Mesh Fairing

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1 Introduction to Mesh Fairing

In 3D geometry processing, raw polygonal surface scans often contain high-frequency measurement noise. Applying isotropic Laplacian smoothing directly to mesh vertex coordinates produces dramatic volume shrinkage and structural collapse over multiple iterations. Screened Poisson surface fairing overcomes this limitation by balancing surface signal smoothing against original geometry preservation. The fairing problem minimizes an energy functional over the 3D surface mesh domain $\mathcal{M}$, formulated as the following continuous partial differential equation (PDE):

$$(I - \alpha\Delta_{\mathcal{M}})u(x, y, z) = u_0(x, y, z) \quad \text{on } \mathcal{M},$$

where u_0 denotes the original noisy 3D vertex positions, u is the smoothed target position field, $\Delta_{\mathcal{M}}$ is the intrinsic Laplace-Beltrami operator defined on the 2D Riemannian manifold $\mathcal{M}$, and the screening parameter $\alpha > 0$ controls the trade-off between spatial smoothing and fidelity to the input model.

2 Discrete Differential Operators

To solve the Screened Poisson equation on an arbitrary 3D triangular mesh $\mathcal{K} = (\mathcal{V}, \mathcal{F})$ with V vertices and F face elements, we discretize the continuous operators using the standard cotangent formula. For a scalar field u defined on the mesh vertices, the discrete cotangent Laplacian matrix $L \in \mathbb{R}^{V\times V}$ is assembled element-wise:

$$L_{i,j} = \begin{cases} \frac{1}{2}(\cot\alpha_{ij} + \cot\beta_{ij}) & \text{if } (i,j) \in \mathcal{E}, \\ -\sum_{k\in\mathcal{N}(i)} L_{i,k} & \text{if } i = j, \\ 0 & \text{otherwise,} \end{cases}$$

where α_{ij} and β_{ij} are the two angles opposite edge (i, j) in adjacent triangles, $\mathcal{E}$ is the set of mesh edges, and $\mathcal{N}(i)$ represents the 1-ring vertex neighbourhood of node i.

To account for non-uniform spatial metric integration across irregular face tessellations, we define the diagonal lumped mass matrix $M \in \mathbb{R}^{V\times V}$ using one-third of the area of all triangles incident to vertex i:

$$M_{i,i} = \frac{1}{3}\sum_{t\in\mathcal{T}(i)} \text{Area}(t),$$

where $\mathcal{T}(i)$ is the set of triangles sharing vertex i. Assembling these operators yields the global discrete Screened Poisson system:

$$(M - \alpha L)u = Mu_0.$$

3 Linear System Properties

The system matrix $A = M - \alpha L$ is sparse, real, and symmetric positive-definite (SPD). Because the mass matrix M is strictly positive on its diagonal and the cotangent Laplacian $-L$ is positive semi-definite , A is guaranteed to be non-singular for any screening parameter $\alpha > 0$.

By leveraging sparse direct Cholesky factorizations ($A = R^T R$), the 3D position fields (x, y, z) are solved concurrently. This formulation enables scalable surface smoothing across complex high-resolution geometries—such as the Stanford Happy Buddha ($543,652$ vertices and $1,087,716$ triangles)—in 30 minutes across parameter sweeps without experiencing volume degradation or numerical instability.